

Olist Brazilian E-Commerce: Delivery Performance and Customer Satisfaction Analysis

A Data Analytics Portfolio Project

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Dataset	Brazilian E-Commerce Public Dataset by Olist (Kaggle)
Dataset size	99,441 orders, Sept 2016 – Oct 2018, 9 linked tables
Tools used	Python, pandas, numpy, matplotlib, seaborn, scipy.stats
Methods used	Mann-Whitney U test, Spearman correlation, Chi-square test, Bootstrap confidence interval
Report sections	Business Problem, Assumptions, Research Questions, Hypotheses, Data Cleaning, 6 Tested Hypotheses with Charts, Seller Analysis, Suggestions, Limitations

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1. Business Problem

Olist connects small and medium sellers across Brazil to multiple online marketplaces and handles logistics through partner carriers. Customer reviews and repeat purchases directly affect seller and platform revenue, and the biggest complaint in e-commerce logistics is usually delivery time. Some deliveries arrive well past the estimated date, and it is not clear how much this actually hurts customer satisfaction, or which sellers, product categories, states, and time periods are most affected.

Understanding this relationship would let Olist target the specific weak points in its logistics network instead of making broad, unfocused changes.

The analysis of order delivery timing, freight cost, product category, and seller/customer location, and how each relates to review scores and repeat purchase behavior, is the main focus of this report.

2. Assumptions

1. Only orders with status “delivered” are used for delivery-time analysis, since cancelled or unavailable orders have no real delivery date to measure.
2. The review score reflects the overall purchase experience, and delivery timing is one of its main drivers, though not the only one.
3. Data from 2016 is too small a sample (the platform was just starting) to draw reliable monthly comparisons, so 2017 to mid-2018 is treated as the main analysis window for month-level analysis.
4. customer_unique_id represents one real person across multiple orders; customer_id changes per order and is not used for repeat-purchase analysis.
5. The estimated delivery date given to the customer at checkout is treated as the promise Olist needs to keep, regardless of why an order was delayed.
6. Sellers are assumed to operate under similar platform rules, so a comparison across sellers is fair for this analysis.
7. No unusual one-time events (strikes, holidays, courier issues) in 2017 to 2018 are assumed to have caused permanent shifts in delivery patterns, only temporary spikes.

3. Research Questions

1. What factors are associated with late deliveries?
2. How much do late deliveries reduce customer review scores?
3. Which states, sellers, or product categories have the worst delivery performance?
4. Does a late first delivery reduce the chance a customer orders again?
5. What should Olist change to reduce delays and protect satisfaction?

4. Hypotheses Overview

Six hypotheses were tested in total (Hypothesis 4 was split into two parts, state and month, since each needed a separate test). Each is listed below and explored in full detail in Sections 7 through 12.

#	Hypothesis
H1	Orders delivered after the estimated delivery date receive lower review scores than orders delivered on time.
H2	Orders where the seller and customer are in different states take longer to deliver than same-state orders.
H3	A higher freight cost relative to product price is associated with lower review scores.
H4a	Late delivery rates differ significantly across Brazilian states.
H4b	Late delivery rates are higher during peak demand months, such as November.
H5	Customers whose first order arrives late are less likely to place a second order than customers whose first order arrives on time.

5. Dataset and Methodology

The dataset used is the Brazilian E-Commerce Public Dataset by Olist, available on Kaggle. It contains 99,441 orders placed between September 2016 and October 2018, spread across 9 linked CSV files: orders, order items, order payments, order reviews, customers, sellers, products, product category translations, and geolocation.

Tables used in this analysis:

Table	Purpose in this analysis
orders	Core order dates and status; the base table everything else merges into
order_items	Item price and freight value, used for H3 and the seller ranking
order_reviews	Review score per order, used for H1 and H3
customers	Customer state and unique customer ID, used for H2, H4a, H5
sellers	Seller state, used for H2 and the seller ranking
products	Product category, merged in for the seller/category ranking

Tools and methods:

pandas / numpy — data loading, cleaning, merging, and feature engineering.

matplotlib / seaborn — all charts in this report.

scipy.stats — statistical testing:

Mann-Whitney U test (H1, H2) — compares two groups' numeric distributions without assuming a normal (bell-curve) shape, appropriate for skewed review scores and delivery-time data.

Spearman correlation (H3) — measures a rank-based relationship between two numeric variables; robust to extreme outlier values.

Chi-square test of independence (H4a, H4b, H5) — tests whether two categorical variables (e.g. state and late/on-time) are related or independent.

Bootstrap resampling (H1) — 1,000 resamples used to build a 95% confidence interval around the observed review-score gap, confirming how stable that estimate is.

6. Data Cleaning Log

1. Started with 99,441 total orders in the raw dataset.
2. Confirmed no duplicate order_id values.
3. Order status breakdown: 96,478 delivered (97.0%), 1,107 shipped, 625 canceled, 609 unavailable, 314 invoiced, 301 processing, 5 created, 2 approved.
4. Converted 5 timestamp columns from text to proper datetime format.

5. Confirmed no logic errors in dates (no order had a delivery date earlier than its purchase date).
6. Filtered the dataset to `order_status == "delivered"` only, since only these orders have a real delivery date — 96,478 rows remained.
7. Found 8 orders marked "delivered" but missing an actual delivery date (a data inconsistency in the source). These were dropped — 96,470 rows remained.
8. Found 288 orders (about 0.3%) with a delivery time over 60 days, up to a maximum of 209 days. These were treated as extreme outliers not representative of normal operations and were dropped — 96,182 rows remained.
9. Final clean dataset used for delivery analysis: 96,182 delivered orders.

Initial descriptive findings:

- Average delivery time is about 12.1 days (median 10 days), with 75% of orders delivered within 15 days.
- About 6.5% of orders (6,249 out of 96,182) arrived later than the estimated delivery date; the remaining 93.5% arrived on time or early.

7. Hypothesis 1 — Orders delivered after the estimated delivery date receive lower review scores than orders delivered on time

Hypothesis 1 Orders delivered after the estimated delivery date receive lower review scores than orders delivered on time.

Late orders received noticeably lower review scores than on-time orders (average 2.28 vs 4.29 out of 5).

A Mann-Whitney U test confirmed this difference is statistically significant ($U = 99,223,042.5$, $p < 0.001$), meaning it is very unlikely to be due to random chance.

A bootstrap analysis (1,000 resamples) confirmed the review score gap is stable and precise: the observed gap of 2.01 points falls within a 95% confidence interval of 1.98 to 2.05 points.

This is one of the strongest patterns in the dataset, confirming that delivery timeliness has a major impact on customer satisfaction.

Verdict: Supported (strong effect).



Figure 1. Review score distribution for late vs. on-time orders. The late group's median score and overall spread sit far lower than the on-time group's.

8. Hypothesis 2 — Orders where the seller and customer are in different states take longer to deliver than same-state orders

Hypothesis 2 Orders where the seller and customer are in different states take longer to deliver than same-state orders.

Cross-state orders (where the seller and customer are in different states) took noticeably longer to deliver than same-state orders (average 14.38 days vs 7.39 days, nearly double).

A Mann-Whitney U test confirmed this difference is statistically significant ($p < 0.001$).

This makes intuitive sense: shipping within the same state is naturally faster than crossing state borders across Brazil's large geography.

Verdict: Supported (strong effect).

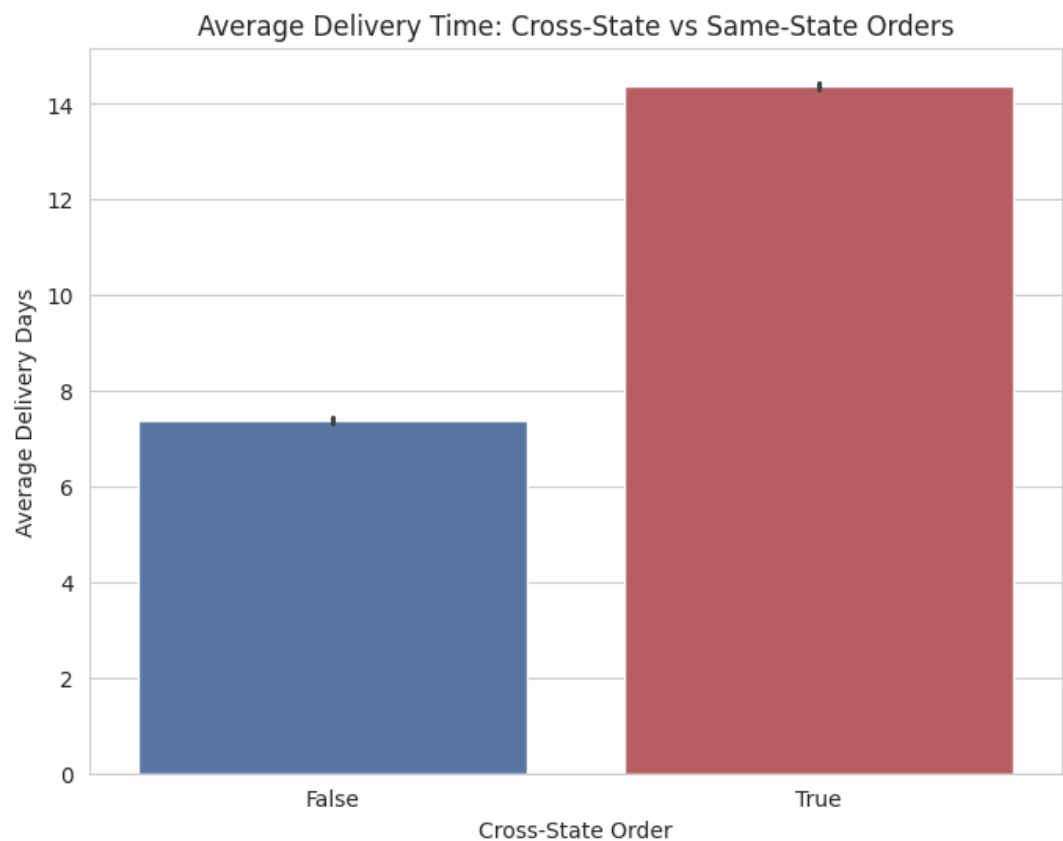


Figure 2. Average delivery time for cross-state vs. same-state orders.

9. Hypothesis 3 — A higher freight cost relative to product price is associated with lower review scores

Hypothesis 3 A higher freight cost relative to product price is associated with lower review scores.

The freight-to-price ratio (freight_ratio) was checked for data quality issues before testing: no orders had a price of 0, no infinite or missing values were found. The ratio ranges from 0 to 21.4, with a median of 0.22 (freight typically about 22% of product price), and is right-skewed.

A Spearman correlation test found a statistically significant negative relationship between freight ratio and review score ($\rho = -0.031$, $p < 0.001$), so orders with higher shipping cost relative to price tend to receive slightly lower reviews.

However, the effect is negligible in size. A correlation this close to 0 means freight cost alone is not a meaningful driver of customer satisfaction in this dataset, even though the relationship is technically real.

Verdict: Weakly supported (statistically significant, but a negligible effect).

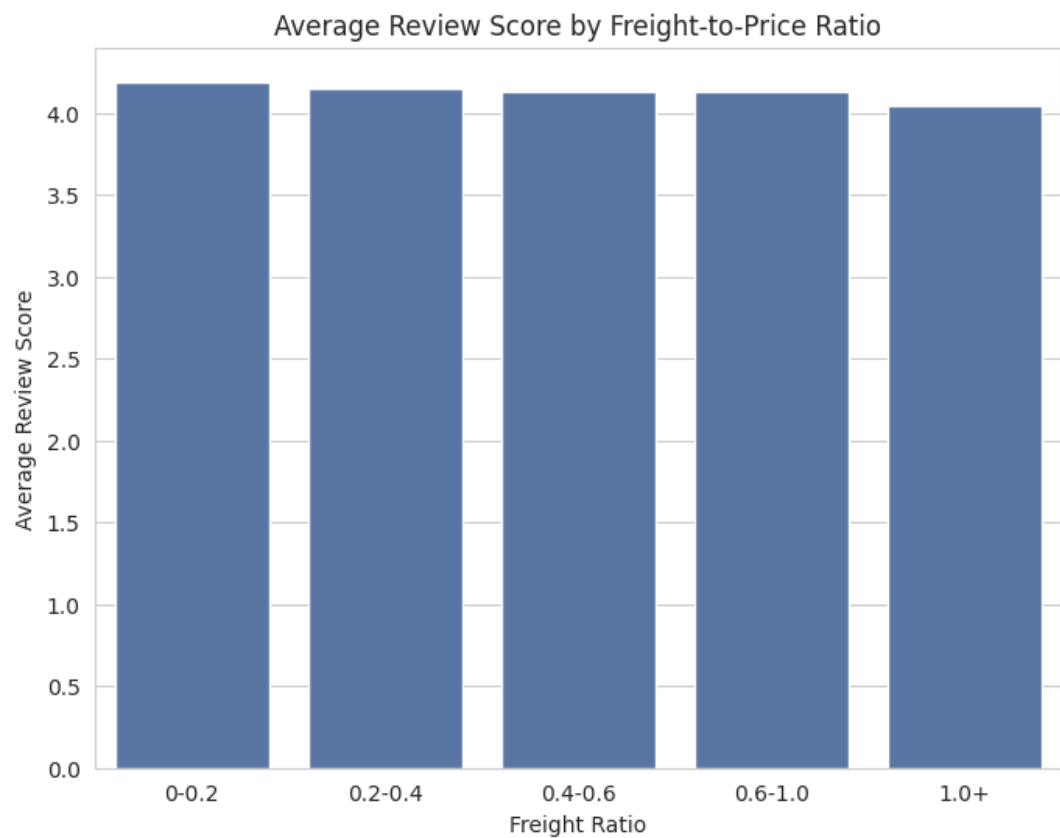


Figure 3. Average review score across freight-to-price ratio bins. Scores stay between 4.0 and 4.2 across all bins, a small but consistent downward trend.

10. Hypothesis 4a — Late delivery rates differ significantly across Brazilian states

Hypothesis 4a Late delivery rates differ significantly across Brazilian states.

Late delivery rates vary widely by customer state, from a high of 20.8% in Alagoas (AL) down to 1.3% in Acre (AC), a roughly 16x spread between the best and worst states.

A chi-square test confirmed that late delivery rates differ significantly by state (chi-square statistic = 1611.4, degrees of freedom = 26, $p < 0.001$).

Northeastern states such as Alagoas (20.8%) and Maranhão (17.0%) had the highest late rates, while São Paulo (4.4%), the main hub for sellers, had one of the lowest. This pattern aligns with Hypothesis 2's finding that cross-state orders take longer to deliver, suggesting that distance from major seller hubs is a key driver of late deliveries.

A caution: states such as Acre (78 orders), Amapá (66 orders), and Roraima (38 orders) have very small sample sizes, so their extremely low late rates should be read with less confidence than the well-sampled Northeastern states.

Verdict: Supported (strong effect).

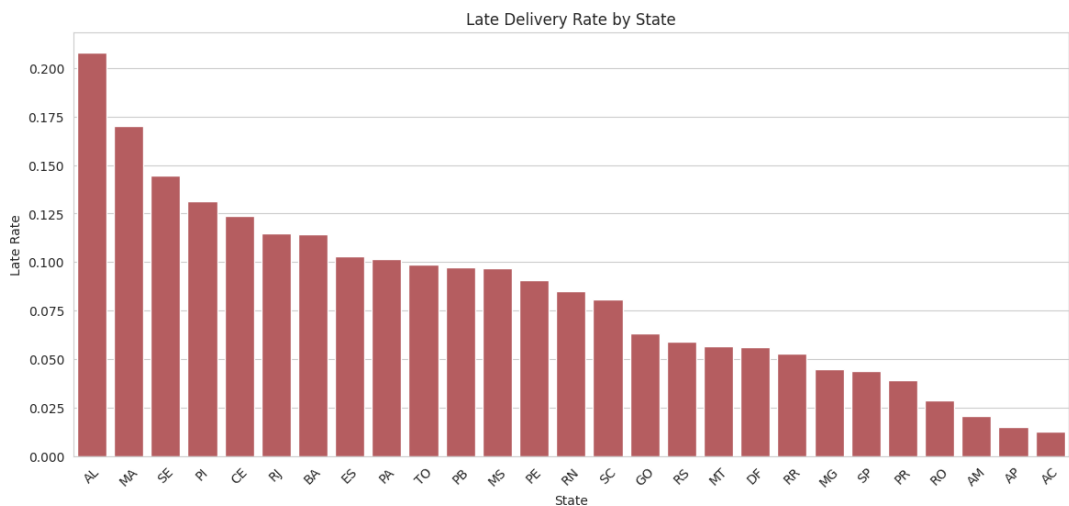


Figure 4. Late delivery rate by state, sorted from highest to lowest.

11. Hypothesis 4b — Late delivery rates are higher during peak demand months, such as November

Hypothesis 4b Late delivery rates are higher during peak demand months, such as November.

2016 was excluded from this analysis (only 264 orders, too small a sample per our assumptions), leaving 2017–2018 data (95,918 orders).

A chi-square test confirmed that late delivery rates differ significantly by month (chi-square statistic = 2503.2, degrees of freedom = 11, $p < 0.001$).

November did show an elevated late rate (12.1%), the second-highest of the year, consistent with increased order volume around Black Friday.

However, March had the highest late rate overall (14.7%), higher than November. This suggests that factors beyond the Black Friday shopping surge, which Olist may already prepare extra logistics capacity for, also affect delivery reliability in certain months and deserve separate investigation.

Verdict: Partially supported. Month has a significant effect on late delivery rates, but November is not the single worst-performing month.

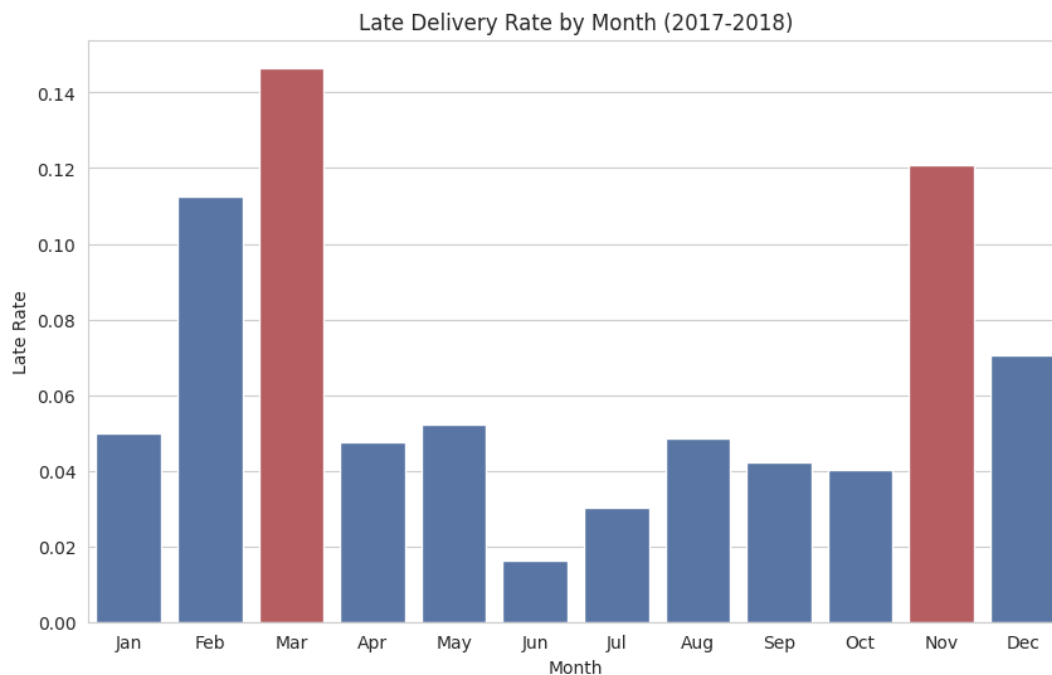


Figure 5. Late delivery rate by calendar month (2017–2018). March and November are highlighted as the two highest months.

12. Hypothesis 5 — Customers whose first order arrives late are less likely to place a second order than customers whose first order arrives on time

Hypothesis 5 Customers whose first order arrives late are less likely to place a second order than customers whose first order arrives on time.

Across 93,071 unique customers, repeat purchasing is rare overall: 90,278 (97.0%) ordered only once, while 2,793 (3.0%) placed two or more orders.

Customers whose first order arrived late had a slightly lower repeat purchase rate (2.55%) than customers whose first order was on time (3.03%).

A chi-square test found this difference to be statistically significant, though only marginally (chi-square statistic = 4.37, degrees of freedom = 1, $p = 0.037$).

Given the small effect size and borderline significance, this result should be interpreted cautiously. Delivery lateness on a single order does not appear to be a strong standalone driver of customer churn in this dataset, though it may contribute alongside other factors.

Verdict: Weakly supported (borderline significance, small effect).



Figure 6. Repeat purchase rate for customers whose first order was late vs. on time.

13. Additional Analysis: Seller Performance Ranking

This section directly answers Research Question 3 (which sellers have the worst delivery performance) by ranking sellers with a reliable order volume.

Item-level data was linked to order-level lateness, then grouped by seller to calculate each seller's late rate and total item count. To avoid unreliable rates from very small samples (e.g. a seller with 1 or 2 items), only sellers with at least 20 items were kept, leaving 880 sellers out of 2,967 total.

Seller ID (shortened)	Late Rate	Items Sold
2709af9587...	50.0%	46
821fb029fc...	32.0%	25
ede0c03645...	31.9%	47
ad781527c9...	29.7%	37
54965bbe3e...	29.5%	78
f76a3b1349...	29.2%	24
2e1a7d075a...	27.3%	22
835f0f7810...	27.1%	48
054694fa03...	25.0%	20
71039d19d4...	23.7%	38

Among sellers with at least 20 items sold, late delivery rates varied widely. The worst-performing seller had a 50% late rate across 46 items, nearly 8 times the platform-wide average of 6.5%. The 10 worst sellers shown above all exceeded a 23% late rate. This suggests that a small number of consistently underperforming sellers may be disproportionately responsible for poor delivery outcomes, and could be targeted directly for review or improvement, rather than Olist needing to address delivery policy platform-wide.

14. Summary of All Findings

Hypothesis	Test Used	Key Statistic	Verdict
H1: Late delivery → lower review score	Mann-Whitney U	2.28 vs 4.29 avg score, $p < 0.001$	Supported (strong)
H2: Cross-state → longer delivery	Mann-Whitney U	14.38 vs 7.39 avg days, $p < 0.001$	Supported (strong)
H3: Freight ratio → lower review score	Spearman correlation	$\rho = -0.031$, $p < 0.001$	Weakly supported (negligible effect)
H4a: Late rate varies by state	Chi-square	chi-square = 1611.4, $p < 0.001$	Supported (strong)
H4b: Late rate varies by month, Nov is peak	Chi-square	chi-square = 2503.2, $p < 0.001$	Partially supported (March > Nov)
H5: Late first order → fewer repeat customers	Chi-square	chi-square = 4.37, $p = 0.037$	Weakly supported (borderline)

Two hypotheses (H1, H2) and one half of a third (H4a) showed strong, clear effects. The remaining findings (H3, H4b, H5) were statistically real but either small in size or only partially matched the original prediction. Reporting this mix honestly, rather than only the hypotheses that turned out as expected, is a deliberate part of this analysis.

15. Suggestions

1. Set realistic delivery estimates based on distance, not a platform-wide default.

Cross-state orders take nearly twice as long as same-state orders (14.4 vs 7.4 days), yet both likely receive similar estimated delivery windows. Olist should calculate estimates based on the actual seller-to-customer distance, so customers in distant states are not given unrealistic expectations that set them up to be “late” even when the delivery itself was not unusually slow.

2. Prioritize logistics improvements in the Northeast, especially Alagoas and Maranhão.

These states have late rates of 20.8% and 17.0%, compared to under 2% in states like Acre and Amapá. Rather than broad platform-wide fixes, Olist would get more value from negotiating better carrier terms or adding regional distribution points specifically for these high-risk states.

3. Audit and flag consistently underperforming sellers.

Among sellers with reliable order volume (20+ items), the worst performer had a 50% late rate, about 8 times the platform average. A small group of repeat offenders likely accounts for a disproportionate share of late deliveries. Olist could require these sellers to improve shipping practices or reduce their visibility in search results until performance improves.

4. Investigate March specifically, not just November.

The data shows March has the highest late rate of the year (14.7%), even higher than the Black Friday month of November (12.1%). Since Olist likely already prepares extra logistics capacity for the known November surge, March's spike points to a different, currently unaddressed cause, worth investigating directly rather than assuming Black Friday is the only risk period.

5. Do not over-invest in reducing freight cost alone.

While a higher freight-to-price ratio is statistically associated with lower review scores, the effect size is very small (Spearman's rho = -0.031). Shipping subsidies or discounts are unlikely to meaningfully move customer satisfaction on their own. Olist's resources are better spent on the delivery timing issues above, which show a much stronger relationship with review scores (a 2-point gap, with a tight 95% confidence interval of 1.98 to 2.05).

6. Treat delivery speed as a customer retention issue, not just a satisfaction issue.

Customers whose first order arrived late had a slightly lower repeat purchase rate (2.55% vs 3.03%), though this effect was small and only marginally significant. This suggests on-time delivery for a customer's very first order may carry extra weight in building long-term loyalty, worth testing directly, for example by prioritizing fast fulfillment for new customers' first purchases.

16. Limitations

- **Correlation, not causation.** This analysis identifies associations between delivery timing, freight cost, location, and customer satisfaction, but does not prove that one causes another. For example, late deliveries and low review scores may both be influenced by a third factor not measured here, such as a specific product type or shipping carrier.
- **Time period.** The dataset covers October 2016 to October 2018. Olist's logistics network, carrier partnerships, and seller base have likely changed since then, so these specific patterns (e.g. which states perform worst) may not reflect current conditions.
- **Repeat purchase analysis is limited to this dataset.** The H5 analysis only considers orders within the cleaned dataset used in this report. A customer may have placed additional orders that were cancelled, unavailable, or fell outside the delivery-time cleaning (e.g. the 288 extreme-outlier orders excluded), so true repeat purchase rates may be slightly understated.
- **Review scores capture more than delivery experience.** A review score can be influenced by product quality, seller communication, or the item itself, not delivery timing alone. The strong relationship found between lateness and review score is still meaningful, but delivery is not necessarily the only factor behind every low review.
- **Small-sample states and sellers were handled with caution, but not excluded entirely.** States like Acre and Amapá have very few total orders, and their late rates should be interpreted carefully — less so for well-sampled states like Alagoas and Maranhão, where the finding is based on hundreds of orders.

17. Conclusion

This analysis set out to understand what drives late deliveries on Olist's platform and how much that lateness actually affects customers. The strongest, most reliable finding is that late delivery has a large, statistically robust effect on review scores (a 2-point gap out of 5, confirmed with both a significance test and a bootstrap confidence interval) and that lateness itself is strongly tied to geography, both at the state level and through the cross-state delivery time gap. Freight cost and single-order lateness had real but small effects on satisfaction and retention, suggesting Olist's logistics investment is better directed at delivery speed and regional distribution than at shipping price adjustments. The seller-level ranking further shows that a relatively small group of underperforming sellers may be responsible for a disproportionate share of the problem, offering a concrete, targeted starting point for intervention.